

An End-to-End Calibration Method for Welding Robot Laser Vision Systems With Deep Reinforcement Learning

Yanbiao Zou and Rui Lan

Abstract—Structured light calibration and robot hand-eye calibration are two of the most crucial parts in welding robot laser vision systems. The main aspect of this calibration is accuracy. To reduce the impact of errors in the calibration process and improve the accuracy, an end-to-end calibration method for laser vision systems based on deep reinforcement learning is proposed. The proposed method involves two dual learning tasks: a pixel-to-point module and a point-to-wrist transformation. The pixel-to-point module predicts the locations of the feature points in the welding image, while the point-to-wrist transformation establishes an accurate conversion relationship from the pixel frame to the wrist frame of the robot. Point-to-wrist transformation consists of two major components: “actor” and “critic.” The “actor” model aims to infer the coordinates of the feature points in the wrist frame of the robot. In offline training, the “critic” model is introduced to guide the learning process of the “actor” model by maintaining the geometric consistency between the image coordinate and the robot coordinate. According to experimental results from the welding robot with continuous motion and changing poses, the proposed method significantly reduces the impact of errors in the calibration process and establishes a more accurate conversion relationship.

Index Terms—Deep reinforcement learning, hand-eye calibration, laser vision system, structured light calibration.

I. INTRODUCTION

WELDING technology is widely used in the automotive sector, engineering machinery, general machinery, and other industries. Recently, welding robots have gradually been replacing workers to address the harsh working environment, highly intensive labor, low efficiency, and other issues associated with these industries. Industrial arc welding robots are usually operated by human operator teaching-programming. To ensure that this working mode is implemented in a specific welding environment, the position of the welding workpiece should initially be located by manual spot welding prior to welding, which causes positioning errors. During the welding process, the workpiece is deformed due to heat, and the actual trajectory deviates from the teaching trajectory, resulting in

low welding precision. The development of machine vision technology has led to the use of welding robots with laser visual detection techniques to correct the playback trajectory for seam tracking. Cameras are the most important sensors in machine vision, which have a wide range of applications in vision measurement [1]–[3]. The laser vision sensor is located in front of the welding torch to measure real-time information of the weld points and guide the welding robot in completing the corresponding movement trajectory. Thus, the welding precision and efficiency can be improved through real-time weld position detection.

Before using the laser vision system to obtain the weld position information, structured light calibration and hand-eye calibration must be performed to establish a conversion relationship from the pixel frame to the wrist frame of the robot. Calibration technology has accelerated the development of robotic communities according to its various usages in sensor calibration and motion sensing [4], and until now, many studies on structured light calibration have been proposed to address particular system configurations. Huynh [5] used four known noncoplanar sets of three collinear world points to compute a unique 4×3 image-to-world transformation matrix. The multiplicity of this matrix and homograph matrix yielded the structured light plane transformation matrix. Zheng and Kong [6] used the intrinsic and extrinsic parameters of the camera to estimate known points on the structured light plane. Then, the homogenous matrix between the camera and structured light was obtained.

With regard to hand-eye calibration, the kernel of the classic method, which was first proposed by Shiu and Ahmed [7] and Tsai and Lenz [8], can be simplified to calculate a matrix equation $AX = XB$. The hand-eye problem has also been applied to multiple sensor calibration including the magnetic/inertial ones [9] and the camera/magnetic ones [10]. A and B , respectively, represent the homogeneous transformation matrix of the end-effector and the homogeneous transformation matrix of the camera between the robot’s two relative positions, and X is the unknown relationship between the robot flange and the sensor. The calibration accuracy will be improved with the increase of calibration data. The matrixes A , B , and X are composed of an orientation rotation matrix R and a translation vector T . Thus, the equation can be divided into two separate sets of equations: a rotation matrix equation and a translation matrix equation

$$R_{Ai}R_X = R_XR_{Bi} \quad (1)$$

$$R_{Ai}T_X + T_{Ai} = R_XR_{Bi} + T_X. \quad (2)$$

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To solve the equation for X , Chou and Kamel [11] used a completely different approach that was based on quaternions and singular-value decomposition analysis. Park and Martin [12] proposed a new Lie theory solution method in the Euclidean group. Daniilidis [13] demonstrated a dual quaternion parametrization, which is an algebraic representation of the screw theory, to describe motions. Strobl and Hirzinger [14] proposed a novel physically based metric on the set of special Euclidean transformation [SE(3)] as the starting point of the maximum likelihood method to optimally estimate within the hand-eye calibration model formulations. The weights of translational and rotational errors can be adapted automatically by the algorithm. Zhao [15] suggested an optimization method that is guaranteed to optimally converge globally based on a second-order cone program convex optimization. Heller *et al.* [16] proposed a pioneering method that removed the requirement for known camera extrinsic parameters from the hand-eye calibration problem. Nguyen and Pham [17] provide a rigorous derivation of the covariance for the hand-eye calibration solution. Wu *et al.* [18] give a universal analytical solution to the hand-eye calibration problem.

A. Motivation

The calibration of the laser vision system in welding robots aims to determine the relationship between the pixel frame and the wrist frame of the robot. To establish the correct conversion relationship, structured light calibration and hand-eye calibration must ensure sufficient accuracy. However, the calibration process is complicated, and the errors of the intermediate process greatly affect the accuracy of the final conversion relationship. Hand-eye calibration has the following differences compared to normal calibration. First, to reduce the arc interference during welding, a filter lens is installed on the front end of the camera lens. Only the light with the same wavelength as the laser wavelength can pass through. Therefore, the position of the calibration point cannot be obtained through the calibration pattern and can be obtained only with the tool center point (TCP). The drawback of this method is that it causes positioning errors. Second, the width of the laser line is 10–12 pixels. The laser plane is calibrated by the centerline of the laser stripe. When the laser shines on the calibration point, the point may deviate from the centerline of the laser stripe, causing sensor detection errors. In addition, the refraction influence of the filter lens cannot be ignored in the continuously changing orientation. All of the above are the main sources of errors in the laser vision system calibration. The errors are both uncertain and unavoidable and directly affect the accuracy of the calibration results. Thus, the focus of this article is to design a more simplified calibration method and establish an accurate conversion relationship.

In the past work, we performed a series of calibration works on the welding robot laser vision system. For camera calibration, we used a 36-mm calibration board [see Fig. 1(a)] and adopted Heikkila's method [19] to estimate the intrinsic parameters. Accurate results were obtained by multiple calibration images and nonlinear optimization. For laser plane calibration,

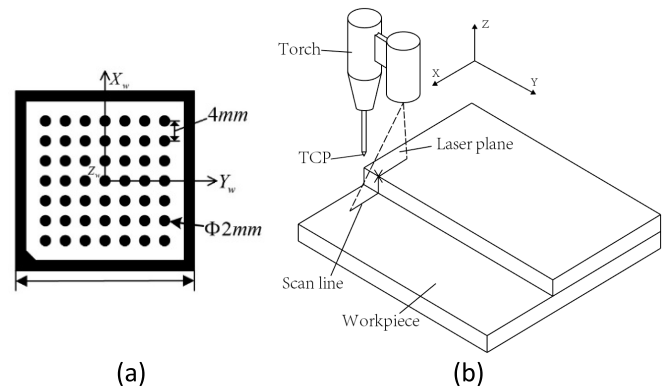


Fig. 1. Laser vision system calibration. (a) 36-mm calibration board. (b) Hand-eye calibration schematic.

we captured the laser weld images and the corresponding calibration board images. Extrinsic parameters and matrix transformation were used to establish the conversion relationship from the pixel frame to the camera frame. The point cloud data were formed by detecting a series of points on the weld image and using the conversion relationship. Then, we used the point cloud data for plane fitting to obtain the equation of the laser plane. With the intrinsic camera parameters and laser plane parameters, we obtained the conversion relationship between the pixel frame and the camera frame through the principle of structured light measurement. For hand-eye calibration, the conversion relationship between the wrist frame of the robot and the sensor measurement frame was used to establish the hand-eye calibration model. As shown in Fig. 1(b), weld images were captured by the laser vision sensor and the weld point pixel coordinates were determined. Then, the position of the weld point in the world frame was obtained by TCP. With multiple sets of measured data, the transformation matrix of the measured frame relative to the wrist frame was obtained by the least squares method. However, due to the small quantity of data and uncertain positioning errors in the above process, it was difficult to obtain an accurate conversion relationship. In [20], semidefinite programming (SDP) was used to identify the direction vectors of the rotation matrix to ensure their orthogonality and reduce the effect of erroneous data. However, the above method analyzes only the calibration results and cannot avoid the effects of intermediate processing errors. Thus, the influence of the calibration errors is still obvious on the robot with changing poses.

To overcome the current algorithmic shortcomings, in this article, we attempt to use neural networks to establish accurate conversion relationships and simplify the calibration process through end-to-end training. Recently, estimating 3-D human poses from monocular RGB imagery has completed the complex conversion of 2D coordinates to 3-D poses. These attempts mainly follow three types of pipelines. The first type [21]–[23] focuses on directly recovering 3-D human poses from 2D input images by utilizing the state-of-the-art 2D pose network architecture to extract 2D pose-aware features with separate techniques and prior knowledge. The second type [24]–[26], [27] concentrates on learning a 2-D-to-3-D pose mapping function. The third type [28]–[30] aims at

integrating the skinned multi-person linear (SMPL) model within a deep network to reconstruct 3-D human pose and shape. Inspired by these studies, this article attempts to detect weld feature points and establish an accurate conversion relationship by maintaining geometric consistency between pixel coordinates and robot base coordinates. Feeding images and labels, our calibration process is performed in an end-to-end manner, which avoids the effects of the above calibration errors.

However, due to insufficient 3-D weld labels, the calibration model cannot be trained in a supervised learning manner. Recent trends [31]–[34] in the reinforcement learning (RL) field combine deep neural networks with RL algorithms by representing RL models, such as value functions or policy [35]. By resorting to deep features, many difficult problems, such as playing Atari games [31] or Go [34], can be successfully solved in semisupervised settings. With the framework of RL, a “critic” model is created to guide the training of the calibration model. Through a semisupervised approach, our calibration algorithm establishes a more accurate conversion relationship.

B. Contributions

Compared with those latest methods, the advantages of our proposed algorithm are mainly manifested in the following two aspects.

- 1) *Improve Calibration Efficiency*: The calibration accuracy will be improved with the increase of calibration data. In [8], [12], [17], [18], and [20], the process of collecting calibration data mainly includes the following steps: the camera is mounted on the robot end-effector, and then the robot transforms different positions and orientations to collect calibration image data from the calibration board shown in Fig. 1(a). Since the quality of the calibration images has a great impact on the calibration results, it can only be identified online by machine vision software, such as Halcon, MATLAB, etc., and then the high-quality calibration images can be saved. It is very difficult to obtain a large amount of data for calibration since the collection process can only be performed manually. In the comparative experiment, Halcon software was used for calibration image collection. The amount of calibration images collected were 279 frames, which took about 3 h. In contrast, the proposed method did not need to collect image data from the calibration plate. The linear angle iron workpieces were selected for the experiment. We taught the robot to scan the workpieces in a changing pose and collected image sequences as well as the corresponding robot end-effector coordinates. Consequently, the calibration data can be obtained. A total of 1500 sets of calibration data were collected during the experiment, which took no more than 10 min. The whole process is automatically completed by program control.
- 2) *Improve Calibration Accuracy*: The application scenario of this article is to use a laser vision system to obtain the position of the welding seam. The calibration process

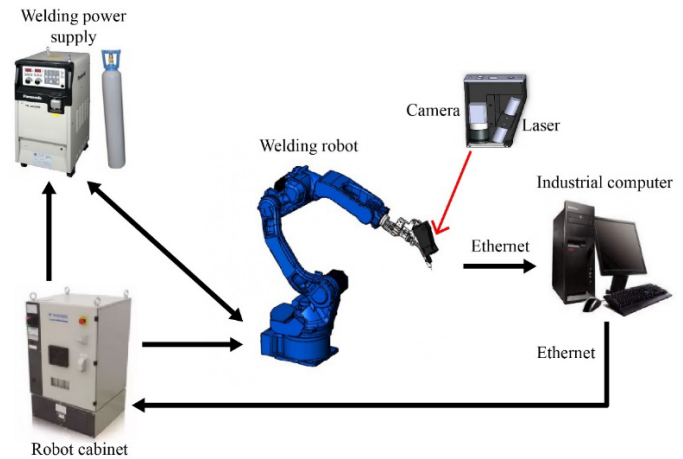


Fig. 2. Automatic welding system.

includes two steps: structured light calibration and hand-eye calibration. The errors in the intermediate process of the two steps will greatly affect the accuracy of the final calibration results. This article proposes an end-to-end calibration method based on RL, which integrates the two steps of structured light calibration and robot hand-eye calibration. The proposed method involves two dual learning tasks, a pixel-to-point module and a point-to-wrist transformation. Taking each image as input, the pixel-to-point module extracts the 2-D pose representations and predicts the locations of the weld points. Point-to-wrist transformation is used to establish the conversion relationship from the pixel frame to the wrist frame of the robot. With the two subnetworks, the calibration process is simplified, and the effects of intermediate process errors are avoided. The network is trained in a semi-supervised manner based on a policy gradient, which effectively solves the insufficient 3-D label problem. Thus, the calibration accuracy can be improved.

This article is organized as follows. In Section II, we briefly describe the welding robot laser vision system and its measurement principle. Section III proposes a calibration method based on RL. End-to-end training simplifies the calibration process and establishes accurate conversion relationships. In Section IV, several experiments are conducted to test the precision of our method. A conclusion drawn from this article is discussed in Section V.

II. AUTOMATIC WELDING SYSTEM

Fig. 2 shows the automatic welding experimental platform. The platform mainly consists of an execution mechanism with six degrees of freedom of motion, welding equipment, laser vision sensor, and an industrial control computer. The laser vision sensor is installed on the welding torch and moves with the torch to detect weld points in real time. The sensor includes a camera and a laser. The laser is mounted at an angle to the camera, and the camera is mounted vertically. The computer obtains the weld seam image captured by the vision sensor and

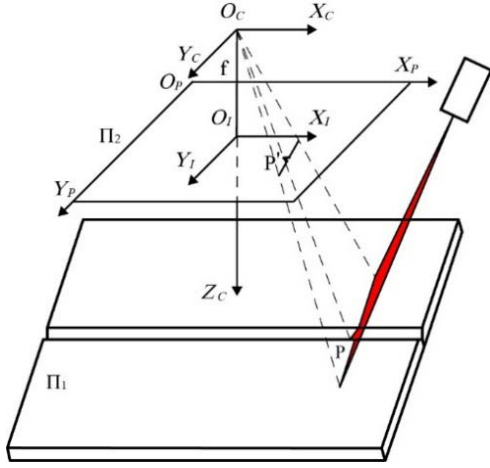


Fig. 3. Perspective projection model.

computes the position of the feature point. According to the position, the robot is driven to obtain the tracking trajectory.

A. Structure Light Measuring System

The measurement principle of our visual system is shown in Fig. 3. Planes Π_1 and Π_2 represent the measurement and imaging planes, respectively. The 3-D coordinates $O_C X_C Y_C Z_C$ denote the camera frame. The 2-D coordinates $O_I X_I Y_I$ denote the imaging frame. $O_P X_P Y_P$ denote the pixel frame. The straight line with $O_C O_I$ is the optic axis of the camera and f is the equivalent focal length. A laser line is formed by the intersection of the laser and the measurement planes. When P is assumed to be a feature point on this laser line, its coordinates are (x, y, z) in the camera frame and (c, r) in the pixel frame. The transformation between these two coordinates is as follows:

$$\begin{aligned} x &= \frac{D(S_x c - C_x S_x)}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \xi} \\ y &= \frac{D(S_y r - C_y S_y)}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \xi} \\ z &= \frac{D f \xi}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \xi} \end{aligned} \quad (3)$$

where $\xi = K C_x^2 S_x^2 - 2 K C_x S_x^2 c + K C_y^2 S_y^2 - 2 K C_y S_y^2 r + S_x^2 c^2 + K S_y^2 r^2 + 1$ and $(S_x, S_y, f, K, C_x, C_y)$ are collectively referred to as the internal parameters of the camera and (A, B, C, D) are the parameters of the laser plane.

The origin O_L of the laser plane frame is obtained by projecting the origin O_C of the camera frame onto the laser plane. In addition, the Z_L axis of the laser plane can be obtained by projecting the Z_C of the camera frame onto the laser plane. The X_L axis of the laser plane can be obtained through the line that passes the O_L point and is perpendicular to Z_L . The Y_L axis, which is perpendicular to the laser plane, can be obtained by the Z_L axis and the X_L axis. The laser plane frame is established, and the point coordinates of the laser plane are expressed as $(x_l, 0, z_l)$. The transformation is

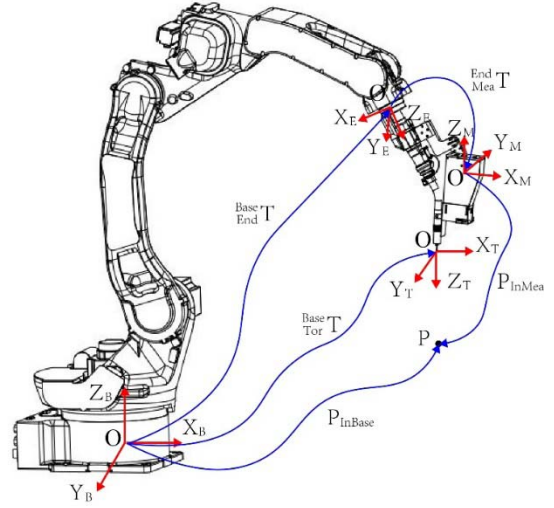


Fig. 4. Arc welding robot coordinate system.

as follows:

$$\begin{bmatrix} x_l \\ 0 \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (4)$$

where the conversion matrix is $_{Camera}^{Mea} T$. With (3) and (4), $(x_l, 0, z_l)$ can be calculated using the pixel coordinates (c, r) .

B. Hand-Eye Calibration Model

Fig. 4 shows the arc welding robot coordinate systems, where $O X_B Y_B Z_B$ is the robot base frame, $O X_E Y_E Z_E$ is the robot wrist frame determined by the TCP, and $O X_T Y_T Z_T$ is the robot tool frame determined by the TCP, and $O X_M Y_M Z_M$ is the measurement frame (laser plane frame). All the points in the work frame can be represented by the robot base frame. $_{Base}^{End} T$ is the transformation matrix of the robot wrist frame relative to the base frame. Similarly, $_{Base}^{Tor} T$ is the transformation matrix of the tool frame relative to the base frame. $_{End}^{Mea} T$ is the transformation matrix of the measurement frame relative to the robot wrist frame.

According to the relationship between the above frames, the equation for the robot hand-eye calibration can be written as:

$$_{Base}^{End} T \times _{End}^{Mea} T \times P_{InMea} = P_{InBase} \quad (5)$$

For a point P in the robot workspace, P_{InMea} is its position matrix in $O X_M Y_M Z_M$, which can be calculated using (4). The position matrix in $O X_B Y_B Z_B$ is P_{InBase} . $_{Base}^{Tor} T$ and $_{End}^{Base} T$ can be obtained by the robot controller calibration algorithm. With $_{Base}^{Tor} T$, P_{InBase} can be obtained by TCP, which is expressed as $[X, Y, Z, 1]^T$. $_{End}^{Base} T^{-1} P_{InBase}$ is a 4×1 matrix and can be rewritten as $[a, b, c, 1]^T$. Then, (5) can be transformed into

$$_{End}^{Mea} T \times P_{InMea} = [a, b, c, 1]^T \quad (6)$$

The above equations are solved by multiple sets of measured data, and we can obtain the transformation matrix $_{End}^{Mea} T$.

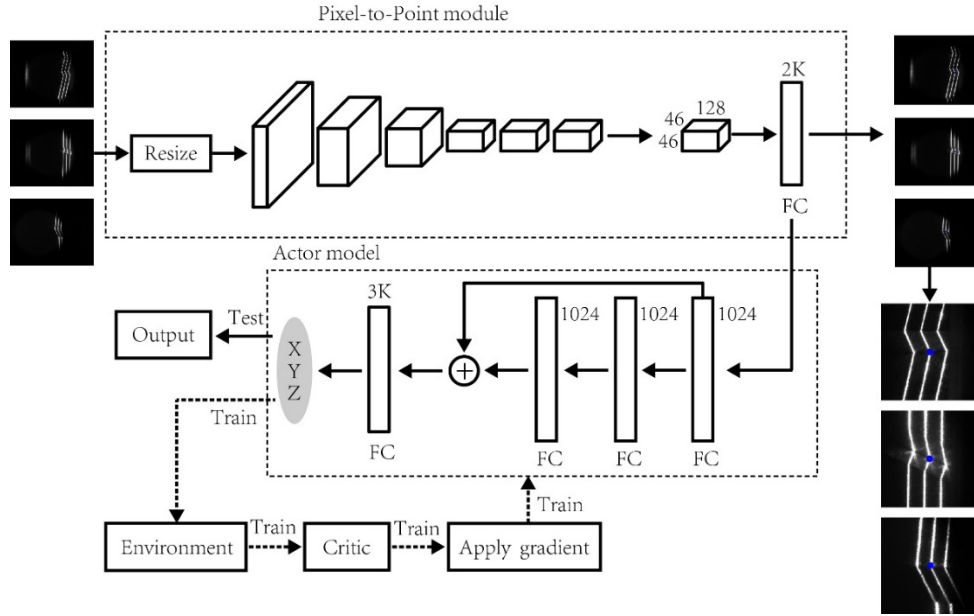


Fig. 5. Proposed laser vision system calibration method. For each monocular weld image, our model predicts the pixel coordinates of the weld points and regresses the 3-D coordinate in the robot wrist frame. The 2-D pixel coordinates are obtained through the pixel-to-point module and then are fed into the “actor” model. In the test phase, the “actor” model directly outputs the 3-D coordinates in the wrist frame. In the training phase, the “actor” model infers the 3-D coordinates of the weld points and interacts with the environment. With the feedback information from the environment, the “critic” model calculates the gradient and applies the gradient to guide the “actor” model to output more accurate 3-D coordinates. The $2K$ and $3K$ denote the dimension of the vector for representing the 2-D pixel coordinates and 3-D wrist coordinates, respectively, formed by K weld points. K is set as 4 in this article.

Combining (4) and (5), the weld point coordinates in the robot base frame can be calculated from its pixel coordinates, and then the robot can be driven to obtain the tracking trajectory.

The transformation relation between the pixel frame and the robot wrist frame can be established through structured light calibration and hand-eye calibration. However, due to the complexity of the calibration process, the errors of the intermediate process, e.g., positioning errors, detection errors, and the refraction influence of the filter lens, are both uncertain and unavoidable. These errors directly affect the accuracy of the calibration results, which ultimately leads to unstable positioning performance of the welding tracking system. To solve these problems, this article proposes an end-to-end calibration method based on RL, which simplifies the calibration process and avoids the influence of the above errors.

III. CALIBRATION ALGORITHM

The proposed method involves two dual learning tasks, a pixel-to-point module and a point-to-wrist transformation. After extracting the position representation, the pixel-to-point subnetwork estimates the pixel coordinates of the weld points for each weld image. The point-to-wrist transformation is used to establish an accurate conversion relationship from the pixel frame to the robot wrist frame. This module is designed to estimate 3-D coordinates in the wrist frame by maintaining geometric consistency. To address the problem of insufficient 3-D labels, the point-to-wrist subnetwork is trained with an RL framework, which involves two major components: “actor” and “critic.” As illustrated in Fig. 5, in the training phase, the “actor” model infers the 3-D coordinates of the weld points in the continuous action space. With the feedback information from the environment, the “critic” model

calculates the policy gradient, which is applied to guide the learning process of the “actor” model. In the test phase, the pixel-to-point module first predicts the pixel coordinates of the weld points. Then, the “actor” model uses the predicted pixel coordinates to regress the 3-D coordinates in the wrist frame. Through end-to-end training, our calibration algorithm simplifies the calibration process and establishes more accurate transformation relations.

A. Pixel-to-Point Module

The objective of the pixel-to-point module is to encode each frame in a given monocular sequence with a compact representation of the position information and to estimate accurate pixel coordinates of weld points. The shallow convolutional layers retain more fine-grained spatial details, which are useful for precise localization and efficient calculation. We build our 2-D module by borrowing the architecture of the convolutional pose machines [36]. See Table I for more details, and a rectified linear unit (ReLU) is employed after each convolutional layer. Note that other state-of-the-art architecture for 2-D pose estimation can also be utilized. As illustrated in Fig. 5, the weld images are first changed to a size of 368×368 pixels and then fed into the 2-D pixel-to-point module as input. The 2-D module outputs the pose-aware feature maps with a size of $128 \times 46 \times 46$ pixels and predicts position vectors with $2K$ entries. The vectors are used to determine the weld point positions.

B. Actor Model

As illustrated in Fig. 5, this module consists of four fully connected layers, and each layer except for the last layer

TABLE I
DETAILS OF THE CONVOLUTIONAL LAYERS IN THE PIXEL-TO-POINT MODULE

	1	2	3	4	5	6
Layer Name	conv1_1	conv1_2	max_1	conv2_1	conv2_2	max_2
Channel (kernel-stride)	64(3-1)	64(3-1)	64(2-2)	128(3-1)	128(3-1)	128(2-2)
	7	8	9	10	11	12
Layer Name	conv3_1	conv3_2	conv3_3	conv3_4	max_3	conv4_1
Channel (kernel-stride)	256(3-1)	256(3-1)	256(3-1)	256(3-1)	256(2-2)	512(3-1)
	13	14	15	16	17	18
Layer Name	conv4_2	conv4_3	conv4_4	conv4_5	conv4_6	conv4_7
Channel (kernel-stride)	512(3-1)	256(3-1)	256(3-1)	256(3-1)	256(3-1)	128(3-1)

contains ReLU operations. The model takes $2K$ position vectors in the pixel-to-point module as the state and outputs the action with $3K$ entries in continuous space, which represents the coordinates of the weld points in the wrist frame. Here, we use $\{x_1^I, y_1^I, \dots, x_K^I, y_K^I\}$ to denote the states and $\{x_1^F, y_1^F, z_1^F, \dots, x_K^F, y_K^F, z_K^F\}$ to denote the actions. Since the 3-D wrist coordinates are directly regressed in the “actor” model and do not need to be calculated by the calibration matrix, the calibration process is simplified, and the intermediate process errors are avoided. Moreover, an identical ResNet [37] mapping is used in this module to quickly pass the information through to avoid overfitting. Therefore, our “actor” model is simple, yet powerful for regression tasks.

C. Critic Model

To facilitate the training of the calibration model, the weld sequences used to construct the environment are collected in advance. For each state, the “actor” model infers the 3-D coordinates of the weld points in the wrist frame and interacts with the environment. After receiving the action signal, the environment outputs the next welding image, which is encoded into the next state by the pixel-to-point module. Thus, in this measurement task, each state is independent. Moreover, the evaluation scheme of the action is mature in this welding localization task. Therefore, we fix the evaluation relationship in the “critic” model to obtain the exact gradient and guide the learning process of the “actor” model. The feedback information from the environment includes the action, the start and end points of the line trajectory, and the actual robot end-effector coordinates corresponding to the currently captured weld image. The feedback actions are decoded from the “actor” model and can be written as $[a_x, a_y, a_z, 1]^T$. The start and endpoints of the line trajectory are $[x_S, y_S, z_S, 0]^T$ and $[x_E, y_E, z_E, 0]^T$, respectively. The current robot end-effector coordinates are used to calculate a 4×4 conversion matrix, which can complete the transformation of point coordinates from the wrist frame to the base frame. Here, we use A_{3d} to denote the feedback actions, M_{end}^{base} to denote the conversion matrixes, P_S to denote the start points, and P_E to denote the endpoints. Then, the evaluation relationship of the “critic” model is given as follows:

$$P_B = \text{Mask} \circ (M_{end}^{base} A_{3d}) \quad (7)$$

where P_B denotes the predicted coordinates of the weld points in the base frame. A mask is used to calculate P_B and can be written as $[1, 1, 1, 0]^T$. $\circ$ denotes the Hadamard product. The line vectors from the start points to the predicted coordinates can be expressed as $V_P = P_B - P_S$, and the actual trajectory vector is expressed as $V_{GT} = P_E - P_S$. With V_P and V_{GT} , the distance from the predicted point to the actual trajectory can be expressed as:

$$D = \left[V_P - \frac{V_P^T V_{GT}}{V_{GT}^T V_{GT}} V_{GT} \right]^T \left[V_P - \frac{V_P^T V_{GT}}{V_{GT}^T V_{GT}} V_{GT} \right]. \quad (8)$$

Combining (7) and (8), an accurate evaluation relationship in the “critic” model can be established, and the deviation distance of the predicted point can be calculated. In our calibration task, the reward is expressed as the positioning accuracy of the predicted weld points. Thus, the reward can be calculated as $\text{reward} = 1/D$. By maximizing the reward signal, the “actor” model can be guided to output more accurate behavior.

D. Training

Our calibration model involves two dual learning tasks, a pixel-to-point module, and a point-to-wrist transformation. For each weld image, the pixel-to-point module predicts the pixel coordinates of the weld points. With the pixel coordinates, the point-to-wrist subnetwork estimates 3-D coordinates in the wrist frame by maintaining geometric consistency. To accelerate the network convergence, the training framework for our calibration model includes supervised learning and RL. Here, we use I_t to denote the t^{th} frame input image, Ψ_R to denote the pixel-to-point module, p_t^{2d} to denote the predicted 2-D pixel coordinates, Ψ_T to denote the “actor” model, and A_t^{3d} to denote the 3-D coordinates in the wrist frame. p_t^{2d} and A_t^{3d} can be calculated as follows:

$$\begin{aligned} p_t^{2d} &= \Psi_R(I_t, w_R) \\ A_t^{3d} &= \Psi_T(p_t^{2d}, w_T) \end{aligned} \quad (9)$$

where w_R, w_T represent the parameters Ψ_R and Ψ_T , respectively.

1) *Training Net With Supervised Learning*: For the 2-D pixel-to-point subnetwork, we build an extra fully connected (FC) layer upon the convolutional layers to generate the $2K$ pixel coordinates. We leverage the Euclidean distances

between the predictions for all K weld points and the corresponding ground truth to train w_R . Formally, we have

$$\min_{w_R} \sum_{t=1}^N \|p_t^{2d}(gt) - \Psi_R(I_t, w_R)\|_2^2. \quad (10)$$

Here, $p_t^{2d}(gt)$ represents the ground truth of the pixel coordinates in the t th frame. To accelerate convergence, normalization is employed for the input and the output.

For the 3-D point-to-wrist transformation, since the action space is large in our calibration task, it is not feasible to directly apply the original RL framework to train our model. Thus, we attempt to improve the training process through supervised learning. First, we use the original 2-D labels and the conversion matrixes from fast calibration to calculate the rough 3-D labels. With the rough 3-D labels, we train the 3-D subnetwork in a supervised learning manner, and the prediction loss is defined as follows:

$$\min_{w_T} \sum_{t=1}^N \|A_t^{3d}(gt^*) - \Psi_T(p_t^{2d}(gt), w_T)\|_2^2 \quad (11)$$

where $A_t^{3d}(gt^*)$ represents the approximate 3-D label obtained by calculating the original 2-D true pixel coordinates. Through the training of rough 3-D labels, our network is unable to establish a satisfactory conversion relationship because the upper limit of accuracy depends on the precision of the calibration matrix. However, the approximate conversion relationship is constructed in the network, which is helpful for the convergence of subsequent training.

After training the above two modules, we fine-tune the whole network to jointly optimize the network parameters $\{w_R, w_T\}$ in a fully end-to-end manner as follows:

$$\min_{\{w_R, w_T\}} \sum_{t=1}^N \|p_t^{2d}(gt) - \Psi_R(I_t, w_R)\|_2^2 + \|A_t^{3d}(gt^*) - \Psi_T(p_t^{2d}, w_T)\|_2^2. \quad (12)$$

2) *Training Net With Reinforcement Learning*: To obtain the 3-D true training data, we need to capture the weld images and determine the pixel coordinates of the weld points as input. Additionally, we drive the robot to the corresponding position of the weld points by teaching-programming and obtain the labels with TCP. Due to difficulties in data collection, it is not advisable to use 3-D labels to train the network in a supervised learning manner. To solve the problem of insufficient 3-D labels, we trained a point-to-wrist subnetwork in an RL framework. First, we accurately locate two points on the linear trajectory through TCP and calculate the real trajectory vector. Then, we teach the robot to move along the linear trajectory with changing poses and capture images in real time. By maximizing the rewards defined in the “critic” model, we can obtain results that are closer to the actual trajectory. Due to the gradient interruption between the “actor” model and the “critic” model, the “actor” model cannot be directly trained by backpropagation. To solve the problem of gradient propagation, the distributed gradient accumulation and asynchronous update methods are used in this article.

We calculate the cumulative gradient in the “critic” model and apply it to guide the “actor” model learning process. In the “actor” model, we use I_i, W_i, B_i to represent the input variables, weight variables, and bias variables of the i th FC layer. D_I^i, D_W^i, D_B^i represent the input, weight, and bias gradient, respectively, in the i -th FC layer. D_{3d}^A represents the output action gradient. In the “critic” model, we use D_{3d}^C to represent the input variables gradient to maximize the rewards and transform it into $3K$ gradient vectors using broadcasting. With D_{3d}^A and D_{3d}^C , we can use the gradient in the “critic” model to guide the learning process of the “actor” model

$$D_{3d} = D_{3d}^A \circ D_{3d}^C \quad (13)$$

where D_{3d} represents the adjusted gradient in the “actor” model. With D_{3d} , we can use backpropagation to calculate the gradient of the variables in the “actor” model and optimize the parameters $\{w_T\}$

$$\begin{aligned} D_I^i &= \begin{cases} D_I^{(i+1)} W_i^T, & i = 1, 3, 4 \\ D_I^{(i+1)} W_i^T + D_I^4, & i = 2 \end{cases} \\ D_W^i &= I_i^T D_I^{(i+1)} \\ D_B^i &= \text{sum}(D_I^{(i+1)}) \end{aligned} \quad (14)$$

where $D_I^5 = D_{3d}$. Through (14), we can adjust the network parameters in the “actor” model to output results that are closer to the actual trajectory.

E. Implementation Details

For weld images, we resize the image region to 368×368 pixels before feeding it into our model. Following the general image preprocessing operation, we transform the input image pixel values into the $[-0.5, 0.5]$ range. To avoid overfitting and accelerate convergence, we augment the training data, including the translation, rotation, interception, and noise. To transform the output of our network into the $[-1, 1]$ range, a max-min normalization strategy is applied. In the testing phase, the predicted 2-D pixel coordinates and 3-D wrist coordinates are transformed into the origin scale according to the maximum and minimum values of the training phase. During the training, the Xavier initialization method [38] is used to initialize the weights of our model. A learning rate of $1e^{-5}$ is employed for training. By feeding images and labels into the two modules, our calibration model is trained in an end-to-end manner to avoid the effects of intermediate processing errors and establish an accurate transformation relationship. With the RL framework, the problem of insufficient 3-D labels is effectively solved by semi-supervised learning.

IV. EXPERIMENTS

The experiment for this calibration method is carried out on the platform shown in Fig. 6. The motion execution mechanism applied is that of the YASKAWA welding robot (model MA1-440), which has six degrees of freedom and a repeat positioning accuracy of ± 0.08 mm. The laser vision sensor, whose measuring accuracy is 0.2 mm, is installed on the welding torch. The robot cabinet (DX200) is used to drive the robot movement and provide the position information to

TABLE II
LASER PARAMETERS

Diode power	Wavelength	Intensity	Distribution	Fan angles	Line uniformity	Bore sighting
100 mW	660 nm	Uniform lengthwise	Gaussian widthwise	30°	Line uniformity to $\pm 5\%$	< 3 mrad

TABLE III
POSITIONING PERFORMANCE OF THE 2-D SUBNETWORK

Cases	Mae X	Var X	$\mu \pm 3\sigma$ (X)	Mae Y	Var Y	$\mu \pm 3\sigma$ (Y)	Time(s)
1 (234)	0.3482	0.0566	[0.18, 0.52]	0.7109	0.235	[0.01, 1.42]	0.037
2 (235)	0.3762	0.0764	[0.15, 0.61]	0.6625	0.1939	[0.08, 1.24]	0.035
3 (235)	0.4248	0.1133	[0.08, 0.76]	0.5043	0.1201	[0.14, 0.86]	0.035
4 (226)	0.3995	0.1009	[0.10, 0.70]	0.8961	0.2562	[0.13, 1.66]	0.037
5 (231)	0.4225	0.1246	[0.05, 0.80]	0.7002	0.2263	[0.02, 1.38]	0.035
6 (236)	0.4275	0.0889	[0.16, 0.69]	0.7137	0.306	[-0.20, 1.63]	0.0371

Note: Time represents the average time of image processing per frame. The results in the table are pixel errors.

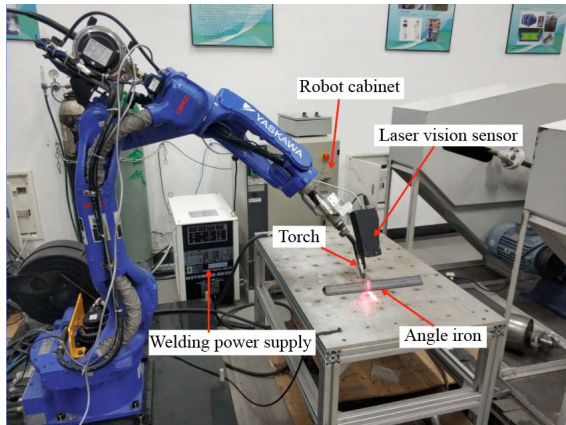


Fig. 6. Experimental platform physical map.

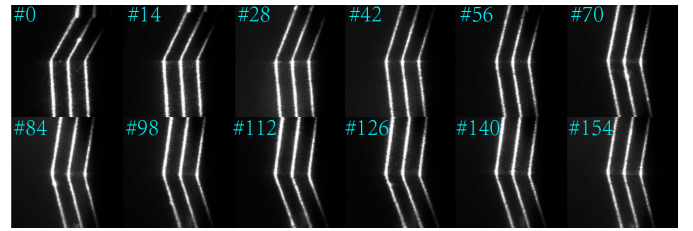


Fig. 7. Weld images. Note: “#N” in the top left corner of the image denotes the Nth frame.

the computer. The image processing module is an industrial computer with a 3.30-GHz Intel i5-4590 quad-core processor, 16-GB RAM, and a single NVIDIA GeForce GTX 1080TI GPU. The optical specifications of the laser used in this article are listed in Table II.

The robot can flexibly move along the linear angle iron workpiece in a wide variety of poses, which enables the laser vision sensor to scan the workpiece in different orientations. To verify the performance of our algorithm in changing orientations, angle iron workpieces are selected for the experiment. First, the actual vectors of the linear trajectory in the base frame are obtained by locating two points on the workpiece with TCP. Then, we teach the robot to scan the workpiece in a changing pose and collect the image sequences as well as the corresponding robot end-effector coordinates. The collected images are fed into our calibration model to output the 3-D coordinates in the wrist frame, while end-effector coordinates are used to calculate the conversion matrix T_{End}^{Base} . With T_{End}^{Base} , the wrist coordinates can be transformed into base coordinates, and the predicted trajectory is obtained. The experiment is carried out in six weld sequences. The obtained images are partially shown in Fig. 7.

Generally, the mean absolute error (Mae) is used to evaluate the static precision of the system, standard deviation/variation (Var) is used to evaluate the stability/robustness of the system, and error is used to evaluate the dynamic precision of the system.

A. Positioning Performance of the 2-D Subnetwork

To evaluate the positioning performance of our algorithm on weld point pixel coordinates, six sets of weld image sequences are fed into the pixel-to-point module. Table III shows the test results. The digits after the case serial numbers indicate the image counts. As seen in Table III, our algorithm can accurately predict the pixel coordinates of weld points in real time, which provides accurate input for establishing conversion relations. In our laser vision system, the width of the laser striped line is about 10 pixels. Therefore, it is considered that the positioning error of the seam feature points should be controlled within 10 pixels to meet the user requirements. Based on the Pauta criterion, the algorithm is stable and robust enough if $6\sigma < 10$ pixels. The calculated results of $[\mu - 3\sigma, \mu + 3\sigma]$ are listed in Table III. 6σ is less than 2 pixels in all cases.

B. Precision and Stability of the 3-D Module

To evaluate the performance of our calibration model, we compare our method with the method in [8], [12], [17], [18], and [20] in precision and stability. First, according to

TABLE IV
COMPARISON OF STATIC PERFORMANCE

Cases	Δ_1		Δ_2		Δ_3		Δ_4		Δ_5		Δ_6	
	Mae(dis)	Var(dis)	Mae(dis)	Var(dis)	Mae(dis)	Var(dis)	Mae(dis)	Var(dis)	Mae(dis)	Var(dis)	Mae(dis)	Var(dis)
1(234)	0.4725	0.0829	1.1470	0.1087	2.0429	0.2001	5.7416	0.1472	2.2107	0.1284	4.9191	0.4741
2(235)	0.8143	0.1863	2.3854	0.3254	3.5607	0.2471	6.7344	0.1507	3.5536	0.1962	2.8879	1.2394
3(235)	0.4803	0.0599	1.6758	0.3997	3.4063	0.4358	5.9836	0.2484	2.9398	0.3402	3.6879	0.9360
4(226)	0.3816	0.0419	1.0643	0.0760	2.0705	0.1291	5.6166	0.0712	2.1465	0.0716	4.8641	0.4166
5(231)	0.4566	0.0987	1.3158	0.0925	2.2549	0.1301	5.9349	0.1041	2.4317	0.1092	4.6518	0.4177
6(236)	0.2881	0.0272	1.1260	0.1069	2.9976	0.1392	5.7563	0.0774	2.5133	0.0989	4.6116	0.2016

Note: Δ_1 represents the result of our method, Δ_2 represents the result with the method in [20], Δ_3 represents the result with the method in [8], Δ_4 represents the result with the method in [12], Δ_5 represents the result with the method in [17], Δ_6 represents the result with the method in [18]

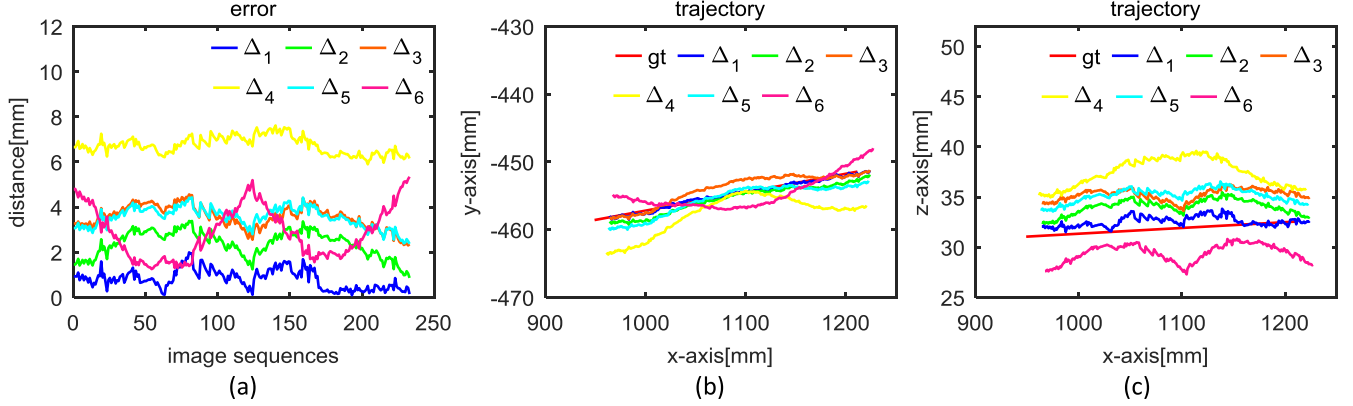


Fig. 8. Experimental results of test set 1. Note: The result of distance error is shown in (a) and the results of the robot trajectory are shown in (b) and (c). “gt” represents the actual trajectory vector.

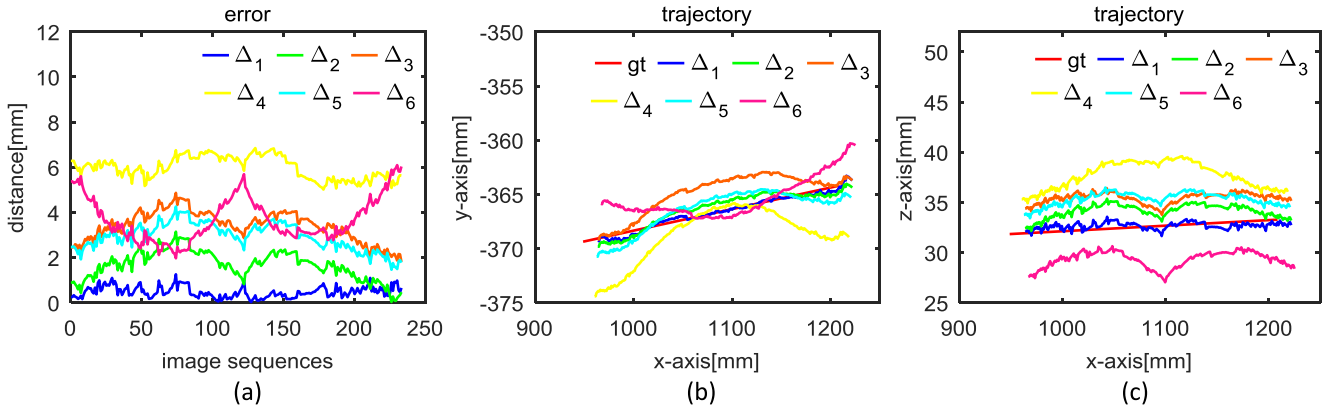


Fig. 9. Experimental results of test set 2. Note: The result of distance error is shown in (a) and the results of the robot trajectory are shown in (b) and (c). “gt” represents the actual trajectory vector.

the algorithm proposed in [8], [12], [17], [18], and [20], 279 high-quality calibration images are reacquired by Halcon software, and the hand-eye calibration conversion matrix is calculated. Second, six sets of weld sequences are fed into a 2-D subnetwork to obtain the pixel coordinates. Third, the pixel coordinates are fed into the calibrated conversion matrix and our “actor” model to output the weld point coordinates in the wrist frame. With T_{End}^{Base} , the base coordinates of the weld points are calculated using all the methods. The distances from the predicted coordinates to the actual trajectory vector are taken as the error, and the results in Table IV are obtained. According to the transformation relationship of our automatic welding system, the scale factor from the image space to the Cartesian space is 0.125 mm/pixel. The width of the laser

striped line is about 10 pixels, which is about 1.25 mm in Cartesian space. Therefore, if the calibration error is less than 1.25 mm, it can meet the application requirements. As shown in Table IV, the average errors of the method Δ_2 are more than 1.25 mm in the cases 2, 3, and 5. Their accuracy cannot fulfill requirements in the continuously changing orientation. In contrast, our proposed algorithm Δ_1 has superior static accuracy and stability.

C. Dynamic Precision of the 3-D Module

We compare our method with the method in [8], [12], [17], [18], and [20] in dynamic precision. The operation processes are consistent with Section IV-B. The experimental results are shown in Figs. 8–10. As seen in Figs. 8–10,

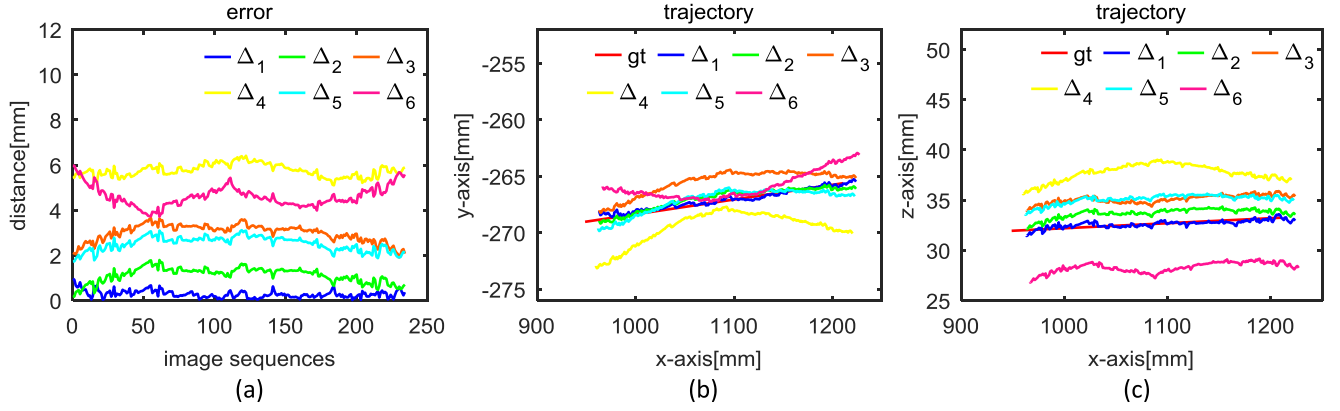


Fig. 10. Experimental results of test set 3. Note: The result of distance error is shown in (a) and the results of the robot trajectory are shown in (b) and (c). “gt” represents the actual trajectory vector.

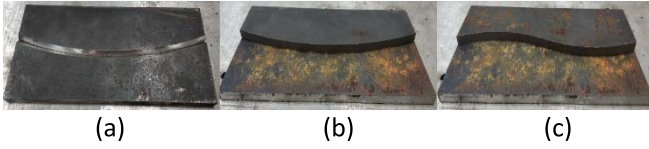


Fig. 11. Welding workpieces. (a) Butt weld seam. (b) Lap weld seam with C-type trajectory. (c) Lap weld seam with S-type trajectory.

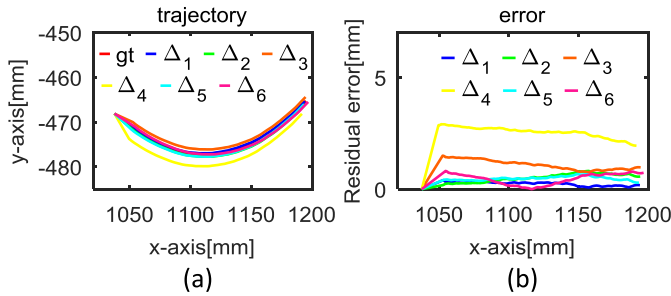


Fig. 12. Coordinates of the torch on the X-O-Y plane (workpiece a). (a) Trajectory. (b) Residual error.

Δ_2 fulfills the requirements in some orientations. However, due to the complex calibration process, the intermediate process error, e.g., the refraction influence of the filter lens and uncertain calibration error, greatly affects the dynamic accuracy in the continuously changing orientation. Δ_3 - Δ_6 can calculate the transformation matrix, but in our laser vision system, it does not have sufficient positioning accuracy and stability. In contrast, our algorithm directly regresses the 3-D wrist coordinates from the “actor” model. Through end-to-end training, the calibration process is simplified, and an accurate conversion relationship is established. Our calibration method has superior dynamic accuracy over other methods.

D. Experimental Verification

To evaluate the performance of our calibration algorithm on actual welding workpieces, we use a butt weld seam [Fig. 11(a)] and a lap weld seam [Fig. 11(b) and (c)], which have arc and curve trajectories. First, the robot is driven to move along the curve seam. The torch tip coordinates of the teaching trajectory are read from the control cabinet, and they are fit into a smooth curve as the theoretical trajectory. For collected images, our calibration model predicts the coordinates of weld points in real time, and the curve obtained is the

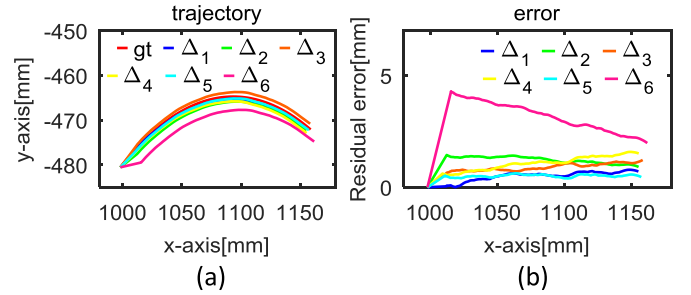


Fig. 13. Coordinates of the torch on the X-O-Y plane (workpiece b). (a) Trajectory. (b) Residual error.

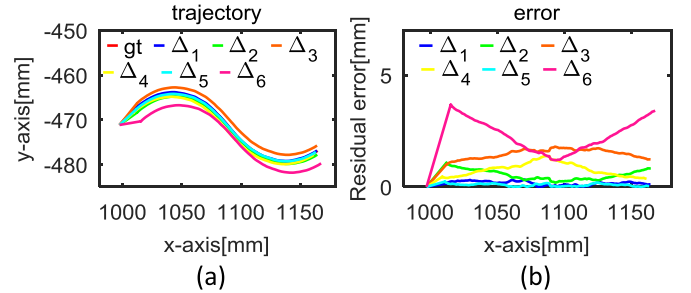


Fig. 14. Coordinates of the torch on the X-O-Y plane (workpiece c). (a) Trajectory. (b) Residual error.

predicted trajectory. To obtain a smooth curve, we adopt a Kalman filter.

The experimental results are shown in Figs. 12–14. As can be seen, when tracking the workpieces (a) and (c), Fig. 11, the positioning error of the previous work [20] is controlled within 1.1 mm, which meets the welding requirement, but the error of the workpiece (b) exceeds 1.25 mm. Among the general methods, Δ_5 has high accuracy and its positioning error is controlled within 1 mm. In contrast, the proposed control method has a control error of less than 0.8 mm, which satisfies the requirements of use and has the best accuracy and stability.

V. CONCLUSION

To avoid the influence of intermediate process errors and to establish an accurate conversion relationship, this article proposes the calibration methods based on deep reinforcement learning for welding robot laser vision systems. The calibration method consists of two stages, i.e., the pixel-to-point network predicts the locations of the feature points in

the welding image, and the point-to-wrist network establishes an accurate conversion relationship from the pixel frame to the wrist frame of the robot. In addition, to handle the issue of insufficient training data set, the reinforcement learning framework is adopted for network training. The comparative tests with the state-of-the-art approaches were performed. The experimental results show that the proposed methods have distinct improvement in static precision, dynamic precision, and stability.

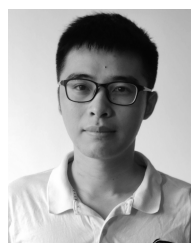
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